



P32

Rules Before Learning: The Robustness Boundary Between Deterministic Predicates and Learned Encoders in Adversarial Safety-Critical Paths

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Working Title (EN): Rules Before Learning: The Robustness Boundary Between Deterministic Predicates and Learned Encoders in Adversarial Safety-Critical Paths

Consensus governance paper P32 • critical flagship of the whole volume • generalizing the H-CSC v2 self-withdrawal lesson into a design principle for UDOS's checkpoints • sister paper to P16 (typed finality), P26 (weighted BFT), and P31 (fail-safe recovery) • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: On the safety-critical path of "deciding whether to trust an Agent product," **do not default to "using a neural network for semantic judgment" being more advanced than "using rules for mechanical checking."** The adversarial experiments of H-CSC v2 prove: the AUROC of the deterministic lexical consistency predicate (0.865–0.982) is significantly better than the learned semantic encoder (0.621–0.744). This paper generalizes this negative result into a design principle—**rules as base, learning as supplement**: all safety-critical checks (P2 stop, P10 handoff, P12 evidence grading) are first based on explainable, reproducible, CI-able rule predicates, and learned models can only "supplement" after the rules have already held the bottom line, and must pass a steelman contrast before entering the safety path.

Evidence scope (stated only once in the whole paper): The core external evidence is H-CSC (arXiv: 2606.07316, v2 retitled Certifiable Semantic Agreement Among LLM Agents: What the Admissibility Instrument Decides); its "lexical predicate AUROC 0.865–0.982 beats learned CRSE encoder 0.621–0.744" and "actively withdrawing honest retention advantage (aggregation distance 5.32°/5.48° vs 4.27°/4.32°, honest retention 0.6475 vs 0.7325)" are all **paper-reported, not independently rechecked**. UDOS-side numbers go back to v7.5.0 P2/P10/P12 (cpu-proto / verified). This paper newly creates no measurements; "rules first" is a design principle, not a contrast experiment already run at all checkpoints. Verification date 2026-09-21.

Abstract

Multi-Agent engineering in 2026 has an intuitive preference: facing questions like "whether two natural-language outputs are semantically consistent" or "whether this handoff is complete," the first reaction is "train/call a semantic encoder to judge"—because neural networks "understand semantics better." The adversarial experiments of H-CSC v2 give a **counterintuitive but key** negative result: in that paper's adversarial scenarios, when deciding whether two texts are semantically consistent, the **simple deterministic lexical consistency predicate** (mechanical comparison by an agreed set of keywords/phrases) has AUROC **0.865–0.982**, while the **learned CRSE semantic encoder** (encoding text into a vector space and then computing similarity) has AUROC only **0.621–0.744**. The tool that "understands semantics better" is instead worse under adversarial conditions.

H-CSC v2 also does one more thing worth recording: after re-checking with a stricter steelman baseline, it **actively withdrew** its own v1 claim that "filtered aggregation retains more honest content"—its core aggregation is 5.32°/5.48° from the honest center, while the steelman baseline is only 4.27°/4.32°; honest retention against steelman M3 is 0.6475 vs 0.7325, and the authors explicitly write "we withdraw it." These two negative results together point to a judgment UDOS must write into its design discipline: **on**

safety-critical paths, a "learned module" is not the default better solution; it must first prove itself, under a fair baseline and under adversarial conditions, to really win over the simplest rule opponent.

This paper generalizes this judgment into the "**rules as base, learning as supplement**" principle and lands it on three existing UDOS safety checkpoints: (1) P2 stop decision—equivocation whole-round invalidation and the $2f+1$ quorum are pure rules, correct; if a "learned semantic-consistency judgment" is introduced to decide the stop level, it must first be based on a rule lexical predicate; (2) P10 TransferBundle—the six-field mechanical check itself is a rule predicate, which is precisely why its 26 contract tests are verified, and do not use a learned model "approximation" to replace six-field non-empty checking; (3) P12 evidence grading—the grading criteria of verified / cpu-proto / cpu-proxy / unverified are rules, and learned models can only give "supplementary hints" after grading is complete, not decide the grading. This paper also draws the boundary of this principle: **rules first is not "rules are omnipotent"**—rule predicates have limited coverage, and for open semantic judgments rules cannot understand, learned models are a necessary supplement; the principle is "rules hold the safety bottom line, learning handles flexibility beyond rules," rather than "use rules for everything."

Keywords: rule predicate; learned encoder; adversarial robustness; steelman baseline; negative result; safety-critical path; evidence grading

Structured Abstract (Background problem → Argument → Evidence → Contributions)

- **Background problem:** Multi-Agent engineering defaults to "neural networks for semantic judgment, rules only for structured fields"; but H-CSC v2's adversarial experiments show that under adversarial conditions the learned semantic encoder is significantly defeated by the simple lexical predicate (AUROC 0.621–0.744 vs 0.865–0.982), and its learned aggregation module actively withdraws its positive claim under the steelman baseline.
 - **Core argument:** Safety-critical paths should be "**rules as base, learning as supplement**"—first use explainable, reproducible, CI-able deterministic rule predicates to hold the bottom line, and learned models only supplement after the rules hold the bottom line, and must pass a steelman contrast before entering the safety path.
 - **Evidence threads:** (1) the lexical predicate vs learned encoder AUROC contrast and active withdrawal of H-CSC v2 (paper-reported); (2) the existing designs of UDOS P2 (equivocation whole-round invalidation as pure rules, verified), P10 (TransferBundle six-field mechanical check, 26 contract tests verified), and P12 (evidence grading as rule criteria).
 - **Contributions:** (i) generalizing H-CSC's single-point negative result into a cross-checkpoint design principle; (ii) landing on the three concrete checkpoints P2/P10/P12 to give "where pure rules are required and where learning can supplement"; (iii) drawing the boundary and falsifiable conditions of "rules first $\neq$ rules omnipotent."
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1 A Default Intuition: Semantic Judgment Should Use Neural Networks

1.1 Where the Intuition Comes From

In LLM engineering there is an almost default judgment: "judging whether two texts are semantically consistent" is a **semantic understanding problem**, and semantic understanding is a neural network's

strength and rules' weakness. So facing such questions, the engineering first reaction is almost always "put on a semantic encoder"—encode the two texts into vectors, compute cosine similarity, and treat them as consistent above a threshold.

This intuition often holds in **non-adversarial** scenarios: within the training distribution, without an attacker deliberately constructing samples that are "literally different but semantically induced consistent" or "literally the same but semantically polluted," the learned encoder indeed understands semantics better than keyword matching.

1.2 But Safety-Critical Paths Are Adversarial Scenarios

The problem is: **the path deciding "whether to trust an Agent product" is precisely the path an attacker most wants to manipulate.** The attacker will not play by the training distribution—he will deliberately construct text where "literal keywords are all correct but semantics are quietly shifted," making the learned encoder (which has seen too many "keyword correct = semantic correct" samples) misjudge as consistent. At this point, the learned model that "understands semantics better," precisely because it has learned too many statistical associations, is less sensitive to samples carefully designed by the attacker.

H-CSC v2's experiments are done precisely in this adversarial scenario. Its conclusion is a reversal of the above intuition.

2 H-CSC v2's Two Negative Results: This Paper's Evidence Base

2.1 The Lexical Predicate Defeats the Learned Encoder

H-CSC v2 (arXiv:2606.07316v2, paper-reported, not independently rechecked) compares under adversarial scenarios two tools for "judging whether two natural-language outputs are semantically consistent":

- **Deterministic lexical consistency predicate:** mechanical, deterministic comparison by an agreed set of keywords/phrases;
- **Learned CRSE semantic encoder:** using a neural network to encode text into a vector space and then compute similarity.

The result (paper-reported numbers): lexical predicate AUROC **0.865–0.982**, learned encoder AUROC **0.621–0.744**. The simple rule tool, under adversarial conditions, is significantly better than the learned model that "understands semantics better."

Why? A reasonable explanation is: the learned encoder's judgment basis is **statistical association**, and the attacker's goal is precisely to break these associations—to decouple "keyword correct" from "semantic correct"; the lexical predicate's judgment basis is **explicitly agreed predicates**, and the attacker either satisfies the predicate or does not, with no "statistically close" fuzzy space. In adversarial games, **rules that are clear, predictable, and precisely computable by the attacker are instead harder to exploit than learned models that are "fuzzy but look smart."**

2.2 Active Withdrawal: Worse Under the Steelman Baseline

H-CSC v2's second negative result is more important than the first, because it is not "someone else's tool failing" but **the authors' self-denial of their own tool**. v1 once claimed its "filtered aggregation" is

better than a simple majority at retaining honest content; after re-checking with a **steelman baseline** (giving the comparison side a stronger, fairer opponent), v2 found (paper-reported):

- Core aggregation distance from the honest center: H-CSC **5.32°/5.48°**, steelman baseline **4.27°/4.32°**—worse;
- Honest retention rate: against steelman M3 **0.6475 vs 0.7325**—worse;
- The authors explicitly: "**we withdraw it.**"

The methodological value of this withdrawal is: it demonstrates **a learned/aggregative module, under a fair baseline, may not only bring no benefit but be worse**. If H-CSC v2 had not done the steelman contrast and directly used v1's "we are better" conclusion, this negative result would never have been recorded. What UDOS learns here is not "don't use learned models" but—**before introducing any learned module, one must first ask: who is its steelman opponent? Did it really win?**

3 Generalization: Rules as Base, Learning as Supplement

3.1 Statement of the Principle

Generalize H-CSC v2's two negative results into one design principle:

Rules as base, Learning as supplement

1. The **bottom line** of safety-critical paths (paths deciding "trust or not, stop or not, accept or not") must be borne by deterministic, explainable, reproducible, CI-able rule predicates;
2. Learned models only, after the rule predicates have already held the bottom line, **supplement** the open semantic judgments rules cannot understand;
3. Any learned module that wants to enter a safety-critical path must first, under a **steelman contrast** and under **adversarial conditions**, prove it is better than the simplest rule opponent; if it cannot prove it, it stays outside the rules.

The reason for this principle is not "learned models are bad" but the **engineering trade-off of auditability and adversarial robustness**: each judgment of a rule predicate can be reviewed by human eyes, re-run by CI, and have its boundary precisely computed by an attacker; learned models' judgments are black boxes whose failure modes under adversarial conditions are hard to predict. The first requirement of a safety-critical path is not "smartest" but "**when something goes wrong it can be found, reproduced, and regressed.**"

Figure 7 The layered protection architecture of rules as base and learning as supplement

Figure 7 The bottom layer is deterministic rule predicates (safety bottom line), the upper layer is learned models filling in, and entering the safety path requires passing the steelman contrast threshold. Conceptual illustration.

3.2 Why Not "Use Rules for Everything"

This principle has an often-misread opposite that must be blocked first: **rules first is not rules omnipotent**. Rule predicates have limited coverage—they can only judge "whether the pre-agreed predicates hold"; for open semantics rules cannot understand (such as "whether this proposal's argument is self-consistent"), learned models are a necessary supplement. The principle is not "ban learned models" but "**put learned models behind the rule defense line, not in front of it.**"

4 Landing on UDOS's Three Checkpoints

4.1 P2 Stop Decision: The Pure-Rule Part Is Correct

The core safety mechanisms of v7.5.0 P2—equivocation whole-round invalidation, $\$n \geq 3f+1$, $\$2f+1$ quorum, and more than $\$f$ silence turning to view change—are all **deterministic rule predicates**. This is precisely why it can be written as 10 CI-able property tests (verified): each is "given a vote combination, the rules mechanically decide."

When the three-level typed finality proposed by P16 (strong_stop / weak_stop / abort) introduces the judgment of "whether the semantic core is consistent," it **must be based on a rule lexical predicate** and cannot directly use a learned semantic encoder—H-CSC v2 has proven the latter has lower AUROC under adversarial conditions. That is: P16's strong_stop judgment should preferentially use "rule comparison of semantic-core fingerprints," and the learned model only acts as a "supplementary hint when rules are uncertain," and must not alone decide the stop level.

4.2 P10 TransferBundle: The Six-Field Mechanical Check Is a Rule Predicate

The TransferBundle handoff contract of v7.5.0 P10 requires six fields non-empty and mechanically checkable, with 26 contract tests verified. This itself is a "rules as base" example—whether a handoff is complete does not rely on "semantically looking clearly passed" but on "all six fields present and formats correct."

P10's discipline should be explicitly written as: **Do not use a learned model "approximation" to replace six-field non-empty checking**. A learned model saying "this handoff is semantically complete" cannot replace the CI-able hard assertion of "all six fields present." Learned models can be used to hint "this handoff, although fields are complete, may semantically have lost context," but they cannot decide "whether the handoff is compliant"—compliance is a rule.

4.3 P12 Evidence Grading: The Grading Criteria Are Rules

v7.5.0 P12 divides evidence into four levels verified / cpu-proto / cpu-proxy / unverified, and the grading criteria are **rules** (whether CI ran, whether single seed, whether cpu-proto). This is the third landing point of rules as base:

- Learned models can be used to "supplement hints"—"this evidence's wording looks like unverified, want to recheck";
- But the grading itself must be rules—verified is precisely "CI ran and ≥ 30 seeds," not accepting "semantically feels reliable."

The reason P12's version-number incident (a silent failure caused by a missing v prefix) is a counterexample is precisely that it **bypassed rule assertions**—if grading were pure rules, "version-number format wrong" should be mechanically intercepted rather than found by human eyes.

5 The Division-of-Labor Table Between Rules and Learning

Table 1 UDOS safety checkpoints: where pure rules and where learning supplements

Checkpoint	Safety bottom line (rule predicate, must be pure rules)	Learned model can fill in (after rules)	Basis
P2 stop	equivocation whole-round invalidation, $\$2f+1\$$ quorum, $\$n\geq 3f+1\$$	hint stop level when rules uncertain	H-CSC lexical predicate AUROC 0.865–0.982 beats learning 0.621–0.744
P16 three-level finality	rule comparison of semantic-core fingerprints	hint strong/weak when rules fuzzy	same; learned aggregation must pass steelman
P10 handoff	TransferBundle six fields non-empty + format	hint "fields complete but semantics may have lost context"	26 contract tests verified
P12 evidence grading	verified/cpu-proto/cpu-proxy/unverified rule criteria	hint "wording suspected unverified, suggest recheck"	version-number incident counterexample
P3 fault governance	three-level circuit-breaking thresholds, kill-switch rules	hint "metrics normal but direction may be drifting"	rules CI-able, reproducible

Figure 8 The rules-and-learning division of labor at UDOS's three checkpoints

Figure 8 The P2/P10/P12 three checkpoints: the safety bottom line is borne by pure rules, and learned models only give fill-in hints. Conceptual illustration.

Table 1's judgment: **All hard judgments "deciding trust or not, stop or not, accept or not" are borne by rule predicates; the learned model's position is "hint," not "decision."**

6 Dross / Failure Boundaries / Falsifiable Conditions

Dross (what not to do):

1. **Do not understand "rules first" as "reject all learned models."** Rules have coverage, and open semantics rules cannot understand must be filled by learned models; this principle is "learning after rules," not "no learning."
2. **Do not put a learned module into the safety path without doing a steelman contrast.** H-CSC v2's withdrawal proves: learned modules may be worse under a fair baseline; going on without contrast amounts to gambling.
3. **Do not confuse "rules explainable" with "rules correct."** Rule predicates themselves may be designed too loosely (misjudging attack samples as pass); rules first requires rules to be covered by adversarial tests, not that writing rules counts.

Failure boundaries:

- When an attacker learns to **precisely mimic the satisfaction conditions of rule predicates** (literally all correct, semantics polluted), rule predicates will likewise be exploited—rules first lowers the risk of "statistical manipulation" but does not eliminate "precisely satisfying rules" attacks;
- Learned encoders in **non-adversarial, within-training-distribution** scenarios may indeed be better than rules; this principle only claims rules first on **safety-critical, adversarial** paths and is not generalized to all semantic tasks;
- H-CSC's AUROC contrast is the result of **that paper's adversarial experiment**, and changing the attack family or predicate definition may give a different conclusion—this principle is "design discipline," not a theorem that "learned models are inferior to rules in all adversarial scenarios."

Falsifiable conditions:

- If in UDOS's adversarial injection experiments the learned semantic encoder's AUROC on stop-level judgment is **significantly and stably higher than** the rule lexical predicate (under a steelman contrast), the hypothesis of "P16 based on lexical predicates" is weakened and needs re-evaluation;
 - If TransferBundle's learned semantic-completeness judgment, beyond the 26 contract tests, can stably detect handoff losses rules miss, the boundary of "six-field pure rules sufficient" needs extension;
 - If evidence grading's learned automatic grading in ≥ 30 -seed reproduction has a lower error rate than rule-based manual grading, "grading must be pure rules" needs loosening.
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7 Conclusion

The most precious legacy H-CSC v2 gives UDOS is not its typed commit design (that is P16's material) but its two negative results: **under adversarial scenarios, the simple lexical predicate defeats the learned semantic encoder; under the steelman baseline, the authors' own learned aggregation module is withdrawn.** These two negative results together are more worth writing into UDOS's design discipline than any positive conclusion—because it corrects an intuition that is defaulted but costs on safety-critical paths: "semantic judgment should use neural networks."

This paper's conclusion is therefore restrained and clear: **rules as base, learning as supplement.** The safety-critical bottom line is borne by rule predicates that are explainable, reproducible, CI-able, and whose boundaries are precisely computable by attackers; learned models fill in after the rules hold the bottom line; any learned module wanting to enter the safety path must first, under a steelman contrast and under adversarial conditions, prove it really wins over the simplest rule opponent. P2's equivocation whole-round invalidation, P10's six-field mechanical check, and P12's evidence grading criteria are already concrete implementations of this principle—any new mechanism introduced by P16/P26/P31 must first pass this gate.

Where deciding "whether to trust an Agent product," the smartest tool is not necessarily the safest tool; a simple rule that can be re-run, reviewed, and CI-regressed is more fit to stand on a safety-critical path than a neural network that "understands semantics better" inside a black box.

References

Classical literature

1. Castro, M., & Liskov, B. (1999). Practical Byzantine Fault Tolerance. OSDI 1999. (the classical skeleton of rule-based majority judgment on safety-critical paths)
2. Vaswani, A., et al. (2017). Attention Is All You Need. arXiv:1706.03762. (the source of learned semantic representation; this paper's argument is not to deny its capability but to limit its position on adversarial safety paths)

2026 frontier materials (existence verified; numbers paper-reported, not independently rechecked)

1. H-CSC author team (2026). Certifiable Semantic Agreement Among LLM Agents: What the Admissibility Instrument Decides (v2). arXiv:2606.07316v2. (under adversarial scenarios the deterministic lexical predicate AUROC 0.865–0.982 beats the learned CRSE encoder 0.621–0.744;

actively withdrawing honest retention advantage under the steelman baseline—aggregation distance 5.32°/5.48° vs 4.27°/4.32°, honest retention 0.6475 vs 0.7325)

UDOS paper volume (evidence grades marked in text)

1. UDOS v7.5.0 P2 "BFT-lite for Multi-Agent Stop Decisions" (equivocation whole-round invalidation, \$2f+1\$ quorum as pure rules, 10 property tests verified).
 2. UDOS v7.5.0 P10 "AgentAsTool and the Measurability of TransferBundle Handoff Information Loss" (six-field mechanical check, 26 contract tests verified).
 3. UDOS v7.5.0 P12 "Evidence-Grading-Driven Reproducible Agent Systems Engineering" (verified / cpu-proto / cpu-proxy / unverified rule grading; version-number incident counterexample).
 4. UDOS v7.6.0 P16 "Typed Finality", P26 "Weighted BFT and Reputation", P31 "Fail-Safe & Recovery" (before introducing new mechanisms, rule-predicate base and steelman contrast are required).
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Evidence Discipline and Reproduction Notes

- This paper is a **design principle** paper with no new experiment; all H-CSC numbers (AUROC intervals, aggregation distance, honest retention rate) are **paper-reported, not independently rechecked**, and are the v2 actively-withdrawn scope, and this paper strictly adopts v2.
- "Rules as base, learning as supplement" is a **design discipline**, not a theorem that "learned models are inferior to rules in all adversarial scenarios"; its failure boundaries (attacker precisely mimicking rules, non-adversarial scenarios, changing attack family) are made clear in Sec. 6.
- The existing rule designs of UDOS P2/P10/P12 are cpu-proto / verified; the "P10 based on lexical predicates, P10 not replaced by learned approximation, P12 grading pure rules" proposed in this paper are all design constraints, whose pros and cons relative to learned approaches need actual measurement under a steelman contrast, marked [RESULT NEEDED] ; ≥ 30 seeds are needed before submission.